Lee–Carter Mortality Modeling: From SVD to Insurance Premiums

A Graduate-Level Bridge Document for the SIMA Project

Covering the k_t Re-estimation Problem, Graduation Compatibility, and the COVID-19 Impact on Mexican Mortality

Andrés González Ortega — SIMA, febrero 2026

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Abstract

This document provides a rigorous, self-contained treatment of the complete mortality modeling pipeline as implemented in the SIMA project: from raw INEGI/CONAPO mortality data through Whittaker–Henderson graduation, Lee–Carter fitting via SVD, k_t projection, and the bridge to actuarial premium calculations.

Special attention is given to a phenomenon encountered with real Mexican data (1990–2024): the k_t re-estimation step fails when b_x has negative components at certain ages, producing a *non-monotone* death residual function that violates the sign-change requirement of Brent's method. We present a complete mathematical analysis of when and why this occurs, prove monotonicity conditions, and document the adaptive solution implemented in the engine.

Every concept is presented in three layers: **formal definition** (blue boxes), **geometric/intuitive explanation** (green boxes), and **practical application with SIMA numerical results** (orange boxes).

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1 Notation and Conventions

Symbol	Domain	Meaning
x	$\{0, 1, \ldots, \omega\}$	Age (integer)
t	$\{t_1, \ldots, t_T\}$	Calendar year
ω	$\mathbb{N}$	Ultimate age (100 in SIMA)
$m_{x,t}$	$\mathbb{R}_{>0}$	Central death rate at age x, year t
$D_{x,t}$	$\mathbb{N}$	Death count (INEGI)
$E_{x,t}$	$\mathbb{R}_{>0}$	Exposure in person-years (CONAPO)
q_x	$[0, 1]$	Probability of death within 1 year
l_x	$\mathbb{R}_{\geq 0}$	Survivors at exact age x
d_x	$\mathbb{R}_{\geq 0}$	Deaths between ages x and $x+1$
a_x	$\mathbb{R}$	Lee–Carter age pattern (log-level)
b_x	$\mathbb{R}$	Lee–Carter age sensitivity
k_t	$\mathbb{R}$	Lee–Carter time index
v	$(0, 1)$	Discount factor $= 1/(1+i)$
D_x, N_x, C_x, M_x	$\mathbb{R}_{>0}$	Commutation functions

Convention: Matrices are denoted in bold ($\boldsymbol{M}$), vectors in bold lowercase ($\boldsymbol{v}$). The log-mortality matrix has dimensions ($n_{\text{ages}} \times n_{\text{years}}$) with ages on rows and years on columns. In SIMA: 101×30 for pre-COVID, 101×35 for full period.

2 The Life Table: Foundation of Actuarial Science

Definition: Life Table

A **life table** is a function $l : \{x_0, x_0 + 1, \ldots, \omega\} \to \mathbb{R}_{\geq 0}$ satisfying:

(i) $l_{x_0} = l_0 > 0$ (the *radix*, typically 100,000),

(ii) $l_x \geq l_{x+1} \geq 0$ for all x (non-increasing),

(iii) $l_{\omega+1} = 0$ (everyone eventually dies).

From l_x we derive all other columns:

$$d_x = l_x - l_{x+1} \qquad \text{(deaths at age } x\text{)} \tag{1}$$

$$q_x = \frac{d_x}{l_x} = 1 - \frac{l_{x+1}}{l_x} \qquad \text{(mortality rate)} \tag{2}$$

$$p_x = 1 - q_x = \frac{l_{x+1}}{l_x} \qquad \text{(survival rate)} \tag{3}$$

At the terminal age: $q_\omega = 1$, $p_\omega = 0$, $d_\omega = l_\omega$.

The Life Table as a Cohort Narrative

Imagine placing 100,000 newborns in a room and watching them age. l_x counts how many remain alive at each birthday. d_x counts the departures each year. q_x is the "exit probability" – at age x, what fraction of those still present will leave before age $x+1$? The fundamental accounting identity holds:

$$\sum_{x=0}^{\omega} d_x = l_0$$

Every member of the initial cohort eventually appears exactly once in some d_x. No one escapes; no one is counted twice.

SIMA: Projected Life Table for Mexico, Year 2029

From the Lee–Carter projection (Section 5), we obtained a projected life table for Mexican mortality in 2029:

Age x	l_x	d_x	q_x	$1000 \cdot q_x$
0	100,000	793	0.00793	7.929
1	99,207	70	0.00071	0.705
30	97,430	139	0.00142	1.422
60	86,649	908	0.01048	10.481
80	51,474	2,799	0.05436	54.359
100	1,859	1,859	1.00000	1000.0

Validation: $\sum d_x = 100{,}000 = l_0$ ✓, $q_{100} = 1.0$ ✓

3 Central Death Rates and the Force of Mortality

Central Death Rate and Force of Mortality

The **central death rate** is the empirical mortality measure:

$$m_{x,t} = \frac{D_{x,t}}{E_{x,t}} \tag{4}$$

where $D_{x,t}$ = observed deaths and $E_{x,t}$ = person-years of exposure at age x in year t.
The **force of mortality** (hazard rate) is the instantaneous rate:

$$\mu(x) = \lim_{h \to 0^+} \frac{q_x(h)}{h} = -\frac{d}{dx} \ln S(x) \tag{5}$$

where $S(x) = \Pr(\text{survive to age } x)$ is the survival function.
Constant force assumption. If μ is constant over $[x, x+1)$, then $\mu_x \approx m_x$ and:

$$p_x = e^{-\mu_x} \approx e^{-m_x} \quad \Longrightarrow \quad \boxed{q_x = 1 - e^{-m_x}} \tag{6}$$

Derivation of $q_x = 1 - e^{-m_x}$. Under constant force μ on $[x, x+1)$:

$$S(x+t) = S(x) \cdot e^{-\mu t} \quad \text{for } t \in [0, 1)$$

The survival probability over one year:

$$p_x = \frac{S(x+1)}{S(x)} = e^{-\mu}$$

The central death rate is:

$$m_x = \frac{\int_0^1 \mu \cdot S(x+t)\, dt}{\int_0^1 S(x+t)\, dt} = \frac{\mu \int_0^1 e^{-\mu t}\, dt}{\int_0^1 e^{-\mu t}\, dt} = \mu$$

Therefore $m_x = \mu_x$, so $p_x = e^{-m_x}$ and $q_x = 1 - e^{-m_x}$. □ □

Why $q_x \neq m_x$

m_x is a *rate* (deaths per person-year), while q_x is a *probability* (chance of dying within the year). For small m_x, they are nearly equal: $q_x \approx m_x$ when $m_x \ll 1$. But for old ages where m_x can exceed 1 (e.g., $m_{100} = 1.78$ in our data), q_x is capped at 1 while m_x is not. The exponential transform $q_x = 1 - e^{-m_x}$ correctly maps $m_x \in (0, \infty)$ to $q_x \in (0, 1)$.

SIMA: The Bridge Method

This conversion is the critical "bridge" in `a09_projection.py`, method `to_life_table()`:

1. Get projected m_x from Lee–Carter: $\hat{m}_x = \exp(\hat{a}_x + \hat{b}_x \hat{k}_t)$
2. Convert: $q_x = 1 - \exp(-\hat{m}_x)$
3. Force $q_\omega = 1$, clip $q_x \in [0, 1]$
4. Build l_x: $l_0 = 100{,}000$, $l_{x+1} = l_x(1 - q_x)$
5. Return `LifeTable` $\rightarrow$ feeds into commutation functions $\rightarrow$ premiums

4 Whittaker–Henderson Graduation

Whittaker–Henderson Penalized Least Squares

Given observed log-rates $\boldsymbol{y} = (\ln m_0, \ln m_1, \ldots, \ln m_n)^\top$, the graduated (smoothed) values $\boldsymbol{z}$ minimize:

$$\mathcal{L}(\boldsymbol{z}) = \underbrace{(\boldsymbol{y} - \boldsymbol{z})^\top \boldsymbol{W} (\boldsymbol{y} - \boldsymbol{z})}_{\text{Fidelity (fit to data)}} + \underbrace{\lambda\, \boldsymbol{z}^\top \boldsymbol{D}_h^\top \boldsymbol{D}_h\, \boldsymbol{z}}_{\text{Roughness penalty}} \tag{7}$$

where:

- $\boldsymbol{W} = \operatorname{diag}(w_0, \ldots, w_n)$ is the weight matrix (typically exposure-based: higher exposure = more reliable data)
- $\lambda > 0$ is the smoothing parameter
- $\boldsymbol{D}_h$ is the h-th order difference matrix. For $h = 2$:

$$\boldsymbol{D}_2 = \begin{pmatrix} 1 & -2 & 1 & 0 & \cdots \\ 0 & 1 & -2 & 1 & \cdots \\ \vdots & & \ddots & & \end{pmatrix} \in \mathbb{R}^{(n-1)\times(n+1)} \tag{8}$$

The closed-form solution follows from $\nabla_{\boldsymbol{z}} \mathcal{L} = \boldsymbol{0}$:

$$\boxed{\hat{\boldsymbol{z}} = \left(\boldsymbol{W} + \lambda\, \boldsymbol{D}_h^\top \boldsymbol{D}_h\right)^{-1} \boldsymbol{W}\, \boldsymbol{y}} \tag{9}$$

Derivation of the solution. Expanding $\mathcal{L}$:

$$\mathcal{L} = \boldsymbol{y}^\top \boldsymbol{W} \boldsymbol{y} - 2\boldsymbol{z}^\top \boldsymbol{W} \boldsymbol{y} + \boldsymbol{z}^\top \boldsymbol{W} \boldsymbol{z} + \lambda\, \boldsymbol{z}^\top \boldsymbol{D}_h^\top \boldsymbol{D}_h \boldsymbol{z}$$

Taking the gradient with respect to $\boldsymbol{z}$:

$$\nabla_{\boldsymbol{z}} \mathcal{L} = -2\boldsymbol{W}\boldsymbol{y} + 2\boldsymbol{W}\boldsymbol{z} + 2\lambda\, \boldsymbol{D}_h^\top \boldsymbol{D}_h \boldsymbol{z} = \boldsymbol{0}$$

Solving:

$$(\boldsymbol{W} + \lambda\, \boldsymbol{D}_h^\top \boldsymbol{D}_h)\boldsymbol{z} = \boldsymbol{W}\boldsymbol{y} \quad \Longrightarrow \quad \hat{\boldsymbol{z}} = (\boldsymbol{W} + \lambda\, \boldsymbol{D}_h^\top \boldsymbol{D}_h)^{-1}\boldsymbol{W}\boldsymbol{y}$$

The matrix $\boldsymbol{W}+\lambda\boldsymbol{D}_h^\top \boldsymbol{D}_h$ is positive definite (sum of PSD matrices with $\boldsymbol{W} \succ 0$), hence invertible. □ □

Graduation as a Tug-of-War

Think of graduation as two forces pulling on the curve:

- **Fidelity** ($\lambda \to 0$): "Stay close to the data!" Result: noisy, jagged rates that perfectly reproduce observations.
- **Smoothness** ($\lambda \to \infty$): "Be as smooth as possible!" Result: a polynomial of degree $h-1$ (for $h=2$, a straight line in log-space).

At $\lambda = 10^5$ (our choice), the penalty strongly favors smoothness while still tracking the major age-pattern features. The $h=2$ penalty targets curvature: it penalizes the second finite difference $\Delta^2 z_x = z_{x+2} - 2z_{x+1} + z_x$, which is zero for linear functions.

SIMA: Log-Space Graduation and the Death-Count Gap

In `a07_graduation.py`, graduation operates in **log-space**: $\boldsymbol{y} = \ln(\boldsymbol{m}_x)$. After solving (9), we exponentiate: $\hat{m}_x^{\text{grad}} = \exp(\hat{z}_x)$.
Critical consequence: Graduated rates do *not* reproduce raw death counts:

$$\sum_x E_{x,t} \cdot \hat{m}_{x,t}^{\text{grad}} \neq \sum_x D_{x,t}$$

SIMA numerical evidence (year 1990):

Quantity	Value
$\sum_x D_{x,t}$ (observed, raw)	472,488
$\sum_x E_{x,t} \cdot m_{x,t}^{\text{raw}}$	472,488
$\sum_x E_{x,t} \cdot \hat{m}_{x,t}^{\text{grad}}$	427,339

Graduation removes the "spiky" features (infant mortality bump, accident hump at young ages) that account for ∼10% of total deaths. This gap is the root cause of the k_t re-estimation failure (Section 6).

Dimension Check

For $n+1$ ages and difference order h:

$$\boldsymbol{y} \in \mathbb{R}^{n+1} \qquad \boldsymbol{W} \in \mathbb{R}^{(n+1)\times(n+1)}$$
$$\boldsymbol{D}_h \in \mathbb{R}^{(n+1-h)\times(n+1)} \qquad \boldsymbol{D}_h^\top \boldsymbol{D}_h \in \mathbb{R}^{(n+1)\times(n+1)}$$
$$\hat{\boldsymbol{z}} \in \mathbb{R}^{n+1}$$

SIMA uses sparse matrices for efficiency: $\boldsymbol{D}_h^\top \boldsymbol{D}_h$ is banded with bandwidth $2h+1 = 5$ for $h=2$.

5 The Lee–Carter Model

Lee–Carter (1992) Log-Bilinear Model

The Lee–Carter model decomposes the log-mortality surface as:

$$\ln(m_{x,t}) = a_x + b_x \cdot k_t + \varepsilon_{x,t} \tag{10}$$

where:

- a_x: average log-mortality at age x (the "shape")
- b_x: age-specific sensitivity to the time trend
- k_t: time index capturing the general mortality level
- $\varepsilon_{x,t} \sim (0, \sigma_\varepsilon^2)$: residual

Identifiability constraints (the model is not unique without these):

$$\sum_x b_x = 1, \qquad \sum_t k_t = 0 \tag{11}$$

What Each Parameter "Does"

- a_x is the "photograph" of mortality – the average death rate pattern across all years. It captures the J-shape: high infant mortality, low childhood rates, rising exponentially at old ages.
- b_x is the "sensitivity dial" – it tells you how much each age responds to overall mortality changes. High b_x at young ages means those ages improved a lot. Negative b_x means that age got *worse* while the nation improved overall.
- k_t is the "volume knob" – one number per year that captures the overall mortality level. When k_t decreases, mortality improves at all ages (proportionally to each age's b_x).

SIMA: Mexican Lee–Carter Parameters

Pre-COVID fit (1990–2019, graduated, no re-estimation):

Metric	Value
Explained variance	77.67%
RMSE (log-space)	0.0618
k_t drift	−1.0764
k_t range	[+23.27, −7.95]

Negative b_x at ages 77, 78, 85 (Mexican data):

Age	b_x	Interpretation
30	+0.0098	Improving with national trend
60	+0.0056	Improving, less sensitive
77	−0.0014	**Worsening** against trend
78	−0.0008	**Worsening** against trend
85	−0.0059	**Most negative** – diabetes/chronic disease

5.1 SVD Estimation Procedure

SVD Fitting of Lee–Carter Parameters

Step 1. Estimate a_x as row means of the log-mortality matrix:

$$\hat{a}_x = \frac{1}{T}\sum_{t=1}^{T} \ln(m_{x,t}) \tag{12}$$

Step 2. Form the residual matrix:

$$R_{x,t} = \ln(m_{x,t}) - \hat{a}_x \tag{13}$$

Step 3. Compute the **Singular Value Decomposition** of $\boldsymbol{R}$:

$$\boldsymbol{R} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^\top \quad \text{where} \quad \boldsymbol{U} \in \mathbb{R}^{n_a \times r},\ \boldsymbol{S} = \operatorname{diag}(\sigma_1, \ldots, \sigma_r),\ \boldsymbol{V} \in \mathbb{R}^{n_y \times r} \tag{14}$$

with $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$ and $r = \min(n_a, n_y)$.
Step 4. Extract the rank-1 approximation:

$$\boldsymbol{R} \approx \sigma_1\, \boldsymbol{u}_1 \boldsymbol{v}_1^\top \tag{15}$$

Set raw parameters: $b_x^{\text{raw}} = u_{x,1}, \quad k_t^{\text{raw}} = \sigma_1 \cdot v_{t,1}$.
Step 5. Apply identifiability constraints:

$$\hat{b}_x = \frac{b_x^{\text{raw}}}{\sum_x b_x^{\text{raw}}} \tag{16}$$

$$\hat{k}_t = k_t^{\text{raw}} \cdot \sum_x b_x^{\text{raw}} - \overline{k^{\text{raw}} \cdot \textstyle\sum_x b_x^{\text{raw}}} \tag{17}$$

Explained variance:

$$\text{EV} = \frac{\sigma_1^2}{\sum_{i=1}^{r} \sigma_i^2} \tag{18}$$

SVD as Finding the Best "Separation of Variables"

The residual matrix $\boldsymbol{R}$ is a surface over (age, year). SVD finds the best way to approximate this surface as a product of an age-function times a year-function: $R_{x,t} \approx b_x \cdot k_t$.
Think of SVD as the matrix equivalent of "factoring" – just as $12 = 3 \times 4$, SVD factors the residual surface into components ranked by importance. The first component (σ_1) captures the dominant pattern; in mortality data, this is the secular improvement trend. The explained variance (77.7% for Mexico) tells you how much of the mortality variation is captured by this single age-year interaction term. The remaining 22.3% is noise, higher-order patterns, or structural features that a single $b_x \cdot k_t$ term cannot capture.

Dimension Check

$$\underbrace{\boldsymbol{R}}_{101\times30} = \underbrace{\boldsymbol{U}}_{101\times30} \cdot \underbrace{\boldsymbol{S}}_{30\times30} \cdot \underbrace{\boldsymbol{V}^\top}_{30\times30}$$

First component extraction: $\boldsymbol{u}_1 \in \mathbb{R}^{101}$ (one value per age), $\boldsymbol{v}_1 \in \mathbb{R}^{30}$ (one value per year), $\sigma_1 \in \mathbb{R}$ (scaling).

6 The k_t Re-estimation Problem

This is the central technical challenge encountered in the SIMA project.

6.1 Why Re-estimate k_t?

The Re-estimation Equation

SVD minimizes error in **log-space**: $\min \|\boldsymbol{R} - \boldsymbol{b}\boldsymbol{k}^\top\|_F^2$. But actuarial applications need accurate **death counts**. Lee & Carter (1992) proposed re-estimating k_t by solving, for each year t:

$$\boxed{\sum_{x=0}^{\omega} E_{x,t} \cdot \exp\big(a_x + b_x \cdot k_t\big) = \sum_{x=0}^{\omega} D_{x,t}} \tag{19}$$

Define:

$$f_t(k) \;=\; \sum_{x=0}^{\omega} E_{x,t} \cdot \exp(a_x + b_x \cdot k) \;-\; \sum_{x=0}^{\omega} D_{x,t} \tag{20}$$

The re-estimated $\hat{k}_t$ satisfies $f_t(\hat{k}_t) = 0$.

6.2 Monotonicity Analysis

Theorem 6.1 (Monotonicity of f_t when $b_x \geq 0$ for all x)**.** *If $b_x \geq 0$ for all x and $b_x > 0$ for at least one x with $E_{x,t} > 0$, then f_t is **strictly increasing**:*

$$f_t'(k) = \sum_{x=0}^{\omega} E_{x,t} \cdot b_x \cdot \exp(a_x + b_x k) > 0 \quad \forall\, k \in \mathbb{R} \tag{21}$$

Moreover:

- $\lim_{k\to-\infty} f_t(k) = -\sum_x D_{x,t} < 0$
- $\lim_{k\to+\infty} f_t(k) = +\infty$

*By the Intermediate Value Theorem, f_t has **exactly one** root.*

Proof. Each summand $g_x(k) = E_{x,t} \cdot \exp(a_x + b_x k)$ satisfies:

$$g_x'(k) = E_{x,t} \cdot b_x \cdot \exp(a_x + b_x k)$$

When $b_x \geq 0$ and $E_{x,t} > 0$, we have $g_x'(k) \geq 0$. Strict inequality holds for any x where $b_x > 0$ and $E_{x,t} > 0$. Since $f_t'(k) = \sum_x g_x'(k)$, we get $f_t'(k) > 0$ for all k.

For the limits: when $k \to -\infty$ and $b_x > 0$, $\exp(a_x + b_x k) \to 0$. When $b_x = 0$, $\exp(a_x + b_x k) = \exp(a_x) > 0$, but the constant terms with $b_x = 0$ are finite, so the sum approaches a finite non-negative value. Thus:

$$\lim_{k\to-\infty} f_t(k) \leq \sum_{x:b_x=0} E_{x,t} \exp(a_x) - \sum_x D_{x,t}$$

In practice, $\sum_x D_{x,t}$ vastly exceeds the constant terms, giving $f_t(k) < 0$ for sufficiently negative k.

Uniqueness follows from strict monotonicity. □ □

Theorem 6.2 (Non-monotonicity when $b_x < 0$ for some x)**.** *If there exists x^* such that $b_{x^*} < 0$ and $E_{x^*,t} > 0$, then f_t is **not** monotonically increasing. Specifically, f_t may have a U-shape with a unique minimum, and the equation $f_t(k) = 0$ may have **zero or two** roots instead of exactly one.*

Proof. The second derivative:

$$f_t''(k) = \sum_{x=0}^{\omega} E_{x,t} \cdot b_x^2 \cdot \exp(a_x + b_x k) > 0$$

since each term $b_x^2 \geq 0$ and at least one is strictly positive. Therefore f_t is **strictly convex**.

A strictly convex function on $\mathbb{R}$ has at most one local minimum and no local maxima. The behavior at the limits:

- $k \to +\infty$: Terms with $b_x > 0$ dominate, so $f_t(k) \to +\infty$.
- $k \to -\infty$: Terms with $b_x < 0$ dominate (since $b_x k \to +\infty$ when $b_x < 0$ and $k \to -\infty$), so $f_t(k) \to +\infty$.

The function approaches $+\infty$ in *both* directions, with a unique minimum k^* where $f_t'(k^*) = 0$. Three cases arise:

1. $f_t(k^*) < 0$: Two roots (one on each side of k^*). Brent's method can find either if bracketed correctly.
2. $f_t(k^*) = 0$: Exactly one root (the minimum touches zero).
3. $f_t(k^*) > 0$: **No roots**. Model-implied deaths are always above observed deaths. The re-estimation equation has no solution.

□ □

The U-Shape: Why Negative b_x Breaks Everything

When all b_x are positive, decreasing k *uniformly decreases* all death rates. Total model deaths drop monotonically – at some point they match the observed total, and we have our root.
But if age 85 has $b_{85} = -0.0059$, then:

$$m_{85}(k) = \exp(a_{85} + b_{85} \cdot k) = \exp(-1.44 - 0.0059k)$$

As k decreases (trying to reduce overall mortality):

- Ages with $b_x > 0$: rates drop (good)
- Age 85 with $b_x < 0$: rate **increases** (negative $\times$ negative = positive exponent)

At $k = -500$, age 85 alone contributes 256,464 model deaths from only 56,618 person-years of exposure. This overwhelms the reduction at all other ages, pulling total model deaths *upward*. The function becomes U-shaped: decreasing at first, hitting a minimum, then increasing again.

SIMA: The Exact Numerical Failure

For Mexican data (1990–2019, graduated), year 1990:

k	Model deaths	$f(k)$	
+200	35,566,695	+35,094,207	
+50	611,018	+138,530	
+20	422,037	−50,451	Root is between 20 and 50
0	353,405	−119,083	
−50	264,017	−208,471	
−200	190,492	−281,996	Minimum region
−500	346,010	−126,478	Curving back UP
−2000	1,815,600,000	$+1.82 \times 10^9$	Age 85 dominates

Brent's method with bracket $[-500, 500]$: $f(-500) = -126{,}478$ and $f(500) = +4.3 \times 10^{12}$. Signs differ – should work!
But the *fallback* bracket $[-2000, 2000]$: $f(-2000) = +1.8 \times 10^9$ and $f(2000) = +2.6 \times 10^{39}$. Both positive – Brent crashes.
The engine now uses **adaptive bracket search**: sample f at 201 points in $[-500, 500]$, find the sign change, bracket tightly. If no sign change exists (Case 3 of Theorem 6.2), use $\arg\min_k |f(k)|$.

6.3 The Graduation–Re-estimation Incompatibility

Proposition 6.3 (Graduation creates an inherent mismatch)**.** *Let $\hat{m}_{x,t}^{grad}$ be the graduated rates. Then in general:*

$$\sum_x E_{x,t} \cdot \hat{m}_{x,t}^{grad} \neq \sum_x D_{x,t} \tag{22}$$

Since $a_x = \frac{1}{T}\sum_t \ln(\hat{m}_{x,t}^{grad})$, the re-estimation equation (19) uses graduated a_x but raw $D_{x,t}$, creating a systematic bias.

Proof. Whittaker–Henderson minimizes (7) in log-space. By Jensen's inequality, for a concave function ln:

$$\mathbb{E}[\ln X] \leq \ln(\mathbb{E}[X])$$

Exponentiating the smoothed log-rates does not generally preserve the sum of death counts:

$$\sum_x E_{x,t} \cdot \exp(\hat{z}_{x,t}) \neq \sum_x E_{x,t} \cdot \exp(\ln m_{x,t}) = \sum_x D_{x,t}$$

In practice, graduation *reduces* total implied deaths (the spiky features at infant ages and old-age accident humps account for excess deaths that get smoothed away). In the SIMA data, the gap is approximately 10%: graduated rates imply ∼427,000 deaths vs. 472,000 observed. □ □

Remark 6.4 (The Principled Choice: Skip Re-estimation for Graduated Data)**.** When fitting Lee–Carter on graduated data, the SVD-estimated k_t minimizes $\|\boldsymbol{R} - \boldsymbol{b}\boldsymbol{k}^\top\|_F^2$, which is *internally consistent* with the log-bilinear formulation. The re-estimation step tries to match raw death counts using graduated parameters – an inconsistent objective. Therefore, `reestimate_kt=False` is the principled choice when using Whittaker–Henderson pre-processing.

7 Brent's Method and Root-Finding

Brent's Method (1973)

Given a continuous function $f : [a, b] \to \mathbb{R}$ with $f(a) \cdot f(b) < 0$ (guaranteed root by the Intermediate Value Theorem), Brent's method combines three strategies:

1. **Bisection**: guaranteed convergence, slow ($O(\log(b - a))$ iterations)
2. **Secant method**: fast (superlinear, order ≈ 1.618), not guaranteed
3. **Inverse quadratic interpolation**: faster when function is smooth

At each step, Brent chooses the fastest method that still maintains a valid bracket. Convergence is guaranteed with superlinear rate.
Requirement: $f(a) \cdot f(b) < 0$. If this fails, `scipy.optimize.brentq` raises `ValueError`.

Why Sign Change is Non-Negotiable

The IVT says: if f is continuous and $f(a) < 0 < f(b)$, then $\exists\, c \in (a, b)$ with $f(c) = 0$. Without a sign change, there is no *guarantee* that a root exists in $[a, b]$, and bisection cannot narrow the interval.
For our $f_t(k)$ in the U-shaped case (Theorem 6.2):

- Bracket $[-500, 500]$: often works (the root lies in $[20, 50]$)
- Bracket $[-2000, 2000]$: fails because $f(-2000) > 0$ and $f(2000) > 0$ (both endpoints are on the same side of the U)

The fix: *adaptively search* for the sign change by evaluating f on a grid, then bracket the root tightly.

8 Mortality Projection: Random Walk with Drift

RWD Model for k_t

After fitting Lee–Carter, project k_t forward as an ARIMA(0,1,0) process with drift:

$$k_{T+h} = k_T + h \cdot c + \sigma \cdot W_h \tag{23}$$

where W_h is a Wiener process (Brownian motion), or in discrete form:

$$k_{T+h} = k_T + h \cdot c + \sigma \sum_{i=1}^{h} Z_i, \quad Z_i \overset{\text{iid}}{\sim} N(0, 1) \tag{24}$$

Parameter estimation from observed $k_1, \ldots, k_T$:

$$\hat{c} = \frac{k_T - k_1}{T - 1} \qquad \text{(drift: average annual change)} \tag{25}$$

$$\hat{\sigma} = \sqrt{\frac{1}{T - 2} \sum_{t=2}^{T} \left(\Delta k_t - \hat{c}\right)^2} \qquad \text{(volatility of innovations)} \tag{26}$$

where $\Delta k_t = k_t - k_{t-1}$.

Drift = Secular Trend, Sigma = Year-to-Year Noise

The drift c is the "average annual improvement" in the mortality index. Negative drift means mortality is improving. The sigma captures how noisy this improvement is year to year.
Stochastic simulations generate many possible futures:

- The **central projection** ($k_T + hc$) is the "expected" path
- Each simulation adds random noise, creating a fan of possibilities
- Quantiles at each horizon give confidence intervals

SIMA: COVID Impact on Drift

Parameter	Pre-COVID	Full (with COVID)	Difference
Drift $\hat{c}$	-1.0764	-0.8548	$+0.2216$
Sigma $\hat{\sigma}$	1.7889	1.5163	-0.2726
Explained var.	77.7%	53.5%	-24.2 pp

COVID makes the drift 0.22 units *less negative* (mortality improvement appears slower). This $\Delta c = +0.22$ propagates through the entire pipeline:

$\Delta k_{T+30} = 30 \times 0.22 = 6.6$ units higher $k_t \implies$ higher mortality $\implies$ 3–10% higher premiums

9 Commutation Functions and the Premium Engine

Commutation Functions

Given a life table $\{l_x\}$ and interest rate i (discount factor $v = 1/(1+i)$):

$$D_x = v^x \cdot l_x \qquad \text{(discounted survivors)} \tag{27}$$

$$N_x = \sum_{j=x}^{\omega} D_j \qquad \text{(sum of future } D\text{'s)} \tag{28}$$

$$C_x = v^{x+1} \cdot d_x \qquad \text{(discounted deaths)} \tag{29}$$

$$M_x = \sum_{j=x}^{\omega} C_j \qquad \text{(sum of future } C\text{'s)} \tag{30}$$

Backward recursion (O(n) computation):

$$N_\omega = D_\omega, \quad N_x = D_x + N_{x+1} \qquad M_\omega = C_\omega, \quad M_x = C_x + M_{x+1} \tag{31}$$

Net Premiums via the Equivalence Principle

The **equivalence principle**: at policy issue, the actuarial present value (APV) of premiums equals the APV of benefits:

$$P \cdot \ddot{a}_{x:\overline{n}|} = SA \cdot A_{x:\overline{n}|} \tag{32}$$

Using commutation functions, D_x cancels in numerator and denominator:

$$\text{Whole life:} \quad P = SA \cdot \frac{M_x}{N_x} \tag{33}$$

$$\text{Term } n\text{:} \quad P = SA \cdot \frac{M_x - M_{x+n}}{N_x - N_{x+n}} \tag{34}$$

$$\text{Endowment:} \quad P = SA \cdot \frac{M_x - M_{x+n} + D_{x+n}}{N_x - N_{x+n}} \tag{35}$$

Why D_x Cancels

The insurance value $A_x = M_x/D_x$ and the annuity value $\ddot{a}_x = N_x/D_x$ both have D_x in the denominator. In the premium equation $P = SA \cdot A_x/\ddot{a}_x$, the D_x cancels:

$$P = SA \cdot \frac{M_x/D_x}{N_x/D_x} = SA \cdot \frac{M_x}{N_x}$$

This is why commutation functions are powerful: they reduce complex actuarial calculations to simple *table lookups and ratios*. No integrals, no summations at premium-calculation time.

SIMA: Premium Impact of COVID

Whole life premiums ($SA = \$1{,}000{,}000$ MXN, $i = 5\%$):

Age	Pre-COVID	Full Period	Δ	% Change
25	\$5,375	\$5,605	+\$230	+4.28%
40	\$10,736	\$11,148	+\$411	+3.83%
60	\$29,829	\$30,796	+\$967	+3.24%

Term 20 premiums are *more* sensitive (8–10% increase) because term insurance is pure mortality risk with no savings component – the drift change hits it harder.

10 Regulatory Validation

Mortality Comparison Metrics

Given projected $\hat{q}_x$ and regulatory benchmark q_x^{reg}:
q_x **ratio** (multiplicative comparison):

$$r_x = \frac{\hat{q}_x}{q_x^{\text{reg}}} \tag{36}$$

RMSE (overall fit over $[x_1, x_2]$):

$$\text{RMSE} = \sqrt{\frac{1}{x_2 - x_1 + 1} \sum_{x=x_1}^{x_2} \left(\hat{q}_x - q_x^{\text{reg}}\right)^2} \tag{37}$$

What the Ratio Tells You

- $r_x > 1$: Projection predicts *higher* mortality than the regulatory table. The projection is more "conservative" (prudent for life insurance pricing).
- $r_x < 1$: Projection predicts *lower* mortality. The regulatory table may be outdated or calibrated to a different population.
- $r_x \approx 1$: Good agreement.

SIMA: Projected vs EMSSA 2009 (Male)

Age	Projected q_x	EMSSA q_x	Ratio
0	0.00793	0.00073	10.86
30	0.00142	0.00145	0.98
50	0.00464	0.00352	1.32
60	0.01048	0.00604	1.74
80	0.05436	0.02095	2.59

Key insight: EMSSA 2009 is calibrated to *insurance company experience* (healthier, wealthier policyholders). Our INEGI-based projection reflects the *general population* – naturally higher mortality, especially at older ages. The divergence (ratios 1.7–2.9 at ages 60+) is expected and actuarially meaningful: it quantifies the selection effect in insured lives.

11 Summary: The Complete Pipeline

Step	Operation	Key Formula	Module
1	Load data	$m_{x,t} = D_{x,t}/E_{x,t}$	`a06`
2	Graduate	$\hat{\boldsymbol{z}} = (\boldsymbol{W} + \lambda \boldsymbol{D}^\top \boldsymbol{D})^{-1} \boldsymbol{W} \boldsymbol{y}$	`a07`
3	Fit Lee–Carter	$\ln m = a_x + b_x k_t$ via SVD	`a08`
4	Project	$k_{T+h} = k_T + hc + \sigma W_h$	`a09`
5	Bridge	$q_x = 1 - e^{-m_x}$, build l_x	`a09`→`a01`
6	Commutations	D_x, N_x, C_x, M_x	`a02`
7	Premiums	$P = SA \cdot M_x/N_x$	`a04`
8	Validate	$r_x = \hat{q}_x/q_x^{\text{reg}}$	`a10`

12 Key Identities and Properties

P1. Death accounting: $\sum_{x=0}^{\omega} d_x = l_0$

P2. Fundamental identity: $A_x + d \cdot \ddot{a}_x = 1$ where $d = iv = i/(1+i)$

P3. SVD optimality: The rank-1 approximation $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^\top$ is the best rank-1 approximation to $\boldsymbol{R}$ in Frobenius norm (Eckart–Young–Mirsky theorem)

P4. Convexity of f_t: $f_t''(k) = \sum_x E_{x,t} b_x^2 e^{a_x + b_x k} > 0$ always (regardless of signs of b_x)

P5. Jensen gap: $\sum_x E_x \exp(\overline{\ln m_x}) \leq \sum_x E_x \cdot \overline{m_x}$ with equality iff all rates are equal

P6. RWD is a martingale (after drift removal): $\mathbb{E}[k_{T+h} - hc \mid k_T] = k_T$

13 What This Document Does Not Cover (Yet)

- **Multi-population models** (Li–Lee, coherent forecasting)
- **CBD model** for old-age mortality (Cairns–Blake–Dowd)
- **Reserve calculations** (prospective method, ${}_tV_x$)
- **Capital requirements** (CNSF stress scenarios, SCR)
- **Model selection**: AIC/BIC for ARIMA order on k_t
- **Goodness-of-fit tests**: residual diagnostics, normality of $\varepsilon_{x,t}$

"All models are wrong, but some are useful."
— George E. P. Box (1976)